Customer Churn Prediction for Subscription-Based Businesses

Project Report

1. Objective & Abstract

Objective

The objective of this project is to build a machine learning model that predicts whether a customer will churn (leave the service) in a subscription-based business. By identifying customers likely to churn, businesses can take proactive retention actions such as targeted promotions, customer support, or loyalty programs.

Abstract

Customer churn is a major challenge for subscription-based businesses because losing customers directly impacts revenue and long-term growth. In this project, customer data is analysed to identify patterns associated with churn behavior. The dataset is cleaned and pre-processed, and new features are engineered to better represent customer behavior.

Two machine learning models — Logistic Regression and Random Forest — are trained to predict churn. The models are evaluated using performance metrics such as Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. The predictions are stored in a CSV file and inserted into a MySQL database for further analysis. A Power BI dashboard is designed to visualize churn insights and help businesses make data-driven decisions.

2. Introduction

Subscription-based companies rely heavily on customer retention to maintain stable revenue. Customer churn refers to the situation when customers stop using a company's service. High churn rates can significantly affect profitability.

Predicting churn allows businesses to take early action to retain valuable customers. Machine learning techniques can analyse historical customer data and identify patterns associated with customers who are likely to leave.

This project uses customer demographic information, subscription details, and usage behavior to build a churn prediction model. The results provide insights into factors influencing churn and help businesses improve customer retention strategies.

3. Goal of the Project

The main goals of this project are:

• Predict whether a customer will churn or stay with the company. • Identify key factors that influence customer churn. • Compare the performance of multiple machine learning models. • Store prediction results in a database for future analysis. • Create a Power BI dashboard to visualize churn insights.

4. Data Overview

The dataset contains customer information collected from a subscription-based service.

Dataset Characteristics:

• Total Records: 350 customers • Total Columns: 9 features

Dataset Columns:

A new feature was created:

Days_Since_Last_Login – number of days since the customer last used the service.

5. Data Cleaning & Preparation

Data preprocessing is an important step to ensure the dataset is suitable for machine learning.

The following steps were performed:

• Converted the Last_Login column into datetime format. • Created a new feature Days_Since_Last_Login to represent customer inactivity. • Converted the Churn column into binary values (Yes = 1, No = 0). • Applied One-Hot Encoding to categorical variables such as Gender and Subscription_Type. • Removed unnecessary columns such as Customer_ID and Last_Login.

These steps helped transform the raw dataset into a format suitable for machine learning models.

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand customer behavior and identify patterns related to churn.

The following analyses were performed:

• Distribution of customer age and monthly charges. • Analysis of subscription types and their relation to churn. • Examination of customer tenure and spending patterns. • Correlation analysis between numerical features.

EDA helped reveal potential relationships between customer characteristics and churn behavior.

7. Machine Learning Models

Two classification models were used to predict customer churn.

Logistic Regression

Logistic Regression is a statistical classification model used for binary outcomes. It estimates the probability that a customer will churn.

Advantages:

• Simple and interpretable • Fast training time • Works well with scaled data

StandardScaler was applied to normalize the feature values before training the model.

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy.

Advantages:

• Handles complex relationships between variables • Reduces overfitting • Provides feature importance information

Random Forest often performs better than simple models for structured datasets.

8. Model Evaluation

The dataset was divided using train-test split:

• Training Data: 80% (280 records) • Testing Data: 20% (70 records)

The models were evaluated using the following metrics:

• Precision • Recall • F1-Score • ROC-AUC Score • Confusion Matrix

Logistic Regression Results

Accuracy: 77%

Confusion Matrix:

ROC-AUC Score: 0.418

Random Forest Results

Accuracy: 74%

Confusion Matrix:

ROC-AUC Score: 0.545

9. Data Storage

The prediction results were saved in two formats:

CSV File

The model predictions were exported to:

churn_predictions_lr_rf.csv

This file can be used for reporting and visualization.

MySQL Database

A MySQL database named churn_db was created. The predictions were inserted into a table called churn_predictions.

This allows the data to be accessed by other applications or dashboards.

10. Power BI Dashboard Design

A Power BI dashboard was designed to visualize churn insights.

Recommended visuals include:

Executive Dashboard

• KPI Cards – Total Customers, Churn Rate, At-Risk Customers • Monthly churn trend

Customer Segmentation

• Churn rate by subscription type • Churn by age group

Behavioral Insights

• Monthly charges vs churn • Tenure months vs churn

Model Insights

• Predicted churn probability distribution • High-risk customers table

Power BI helps transform the model output into actionable business insights.

11. Insights & Recommendations

Based on the analysis, the following insights were observed:

• Customers with longer inactivity periods are more likely to churn. • Subscription type and monthly charges influence churn behavior. • Customers with shorter tenure tend to churn more frequently.

Recommended actions:

High-risk customers (probability > 0.7): • Personalized retention campaigns • Discount offers or loyalty rewards

Medium-risk customers: • Email engagement campaigns • Feature updates or product education

Low-risk customers: • Upsell premium services

12. Future Improvements

The project can be further improved by:

• Increasing dataset size for better model accuracy. • Adding additional behavioral features such as login frequency. • Testing advanced models such as XGBoost or Neural Networks. • Implementing real-time churn prediction systems. • Automating model retraining and monitoring.

13. Conclusion

This project successfully developed a machine learning model to predict customer churn for a subscription-based business. The dataset was cleaned and processed, and meaningful features were engineered to capture customer behavior.

Two classification models — Logistic Regression and Random Forest — were trained and evaluated. The results were stored in both CSV and MySQL formats for further analysis. A Power BI dashboard design was proposed to visualize churn trends and support business decision-making.

By implementing such predictive models, businesses can proactively identify customers at risk of leaving and implement targeted retention strategies to improve customer loyalty and revenue.